

Multimodal Natural Language Processing

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1. Motivation and the grounding problem
2. Learning Outcomes
3. Vision-Language Models
 1. The task space
 2. CLIP, the contrastive objective, and modern vision encoders
 3. The Visual BERT era
 4. Multimodal LLMs, visual instruction tuning, and visual tokens
4. Multimodal LLMs in 2025–2026
 1. Open-weight and frontier flagships, and how they are trained
 2. Native multimodality, unified generation, and omni models
 3. Video, documents, retrieval, GUIs, and robots
 4. Efficiency, benchmarks, and failure modes
5. Multimodal Architectures
6. Retrieval-Augmented Generation (RAG)

- ▶ Traditional NLP models are **unimodal**: text in, text out; they can neither see nor hear the world their language describes.
- ▶ Human communication is grounded in perception: much of what language refers to arrives through vision, sound, and the other senses.
- ▶ **Vision-language models** align images with text at web scale, and today's assistants are **multimodal LLMs** built on top of that alignment.
- ▶ This lecture builds from contrastive alignment (CLIP and its successors) to the projector-and-instruction recipe behind multimodal assistants, and on to the natively multimodal and omni models of 2025–2026.

multimodal adjective

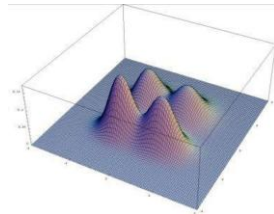
[muhl-tee-moh-duhl]

: having or involving several modes, modalities, or maxima

multimodal distributions

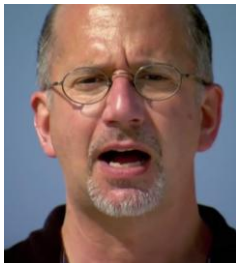
multimodal therapy

In our case, focusing on NLP: text + one or more other *modality* (images, speech, audio, olfaction, others). We'll mostly focus on images as the other modality.



A multimodal distribution: several maxima.
Definition: Merriam-Webster,
"multimodal".

- ▶ **McGurk effect** (McGurk & MacDonald, 1976)
 - Classic demonstration of interaction between modalities (audio + vision) in perception
 - Speech sound is misheard depending on visual (lip movement) cues
- ▶ [Watch this demonstration on YouTube](#)



Source: frame from “McGurk effect: Auditory Illusion”, BBC *Horizon*.

- 2015** Captioning and VQA become benchmark tasks: Show and Tell (CVPR 2015), VQA (ICCV 2015)
- 2019** BERT-style vision-language pretraining: ViLBERT (NeurIPS 2019), LXMERT (EMNLP 2019)
- 2021** Contrastive alignment at web scale: CLIP (ICML 2021)
- 2022** Few-shot multimodal in-context learning: Flamingo (NeurIPS 2022)
- 2023** Visual instruction tuning (LLaVA); frontier chatbots gain vision (GPT-4V system card, Sep 2023)
- 2024** Natively multimodal frontier models: GPT-4o, one network end-to-end across text, vision and audio
- 2025** Images enter the chain of thought (o3); open omni models; native image generation
- 2026** Native multimodality becomes the default in frontier and open releases alike

Every milestone above is cited in the references.

- ▶ Explain why language needs grounding, and map the vision-language task space to its canonical benchmarks.
- ▶ Explain contrastive vision-language alignment (CLIP, SigLIP) and what dual encoders can and cannot do.
- ▶ Describe visual instruction tuning, the projector recipe behind multimodal assistants, and how images become tokens.
- ▶ Survey the 2025–2026 generation: open-weight flagships, natively multimodal frontier systems, unified understanding-and-generation models, and omni models.
- ▶ Explain the applications beyond chat: video, documents, multimodal retrieval, GUI agents, and vision-language-action robotics.
- ▶ Interpret multimodal benchmarks (MMMU, MMMU-Pro) and identify standing failure modes such as object hallucination, visual token cost, and multimodal jailbreaks.
- ▶ Compare early, middle and late fusion, and the classic challenges of multimodal integration.

Vision-Language Models

Parts of this section adapted from D. Kiela, “Multimodal Deep Learning (with an NLP focus)”,
guest lecture, Stanford CS224n, Winter 2023.

- **Visual question answering:** free-form questions about an image (VQA, ICCV 2015). VQAv2 (CVPR 2017) pairs every question with two images that yield two different answers, so language priors alone cannot score well.
- **Captioning:** describe the image (COCO Captions: five human captions per image, CIDEr and BLEU scoring).
- **Referring expressions:** localize “the man in the red hat” (RefCOCO family, 2016).
- **Reading text in images:** TextVQA (CVPR 2019).
- **Documents and charts:** DocVQA (WACV 2021), ChartQA (ACL Findings 2022).



VQAv2's complementary pairs: same question, two images, two answers. Source: Goyal et al., “Making the V in VQA Matter”, CVPR 2017, [arXiv:1612.00837](https://arxiv.org/abs/1612.00837), Figure 1.

Multimodal LLMs are still evaluated on descendants of all five tasks.



- ▶ Developed by OpenAI (Radford et al., 2021)
- ▶ Trained on 400M (image, text) pairs
- ▶ Learns a joint embedding space using contrastive loss
- ▶ Image and text encoders are trained to match correct pairs

Use cases:

- ▶ Zero-shot classification
- ▶ Semantic similarity
- ▶ Prompt-based image querying



Architecture:

- ▶ ResNet / ViT for images
- ▶ Transformer for text
- ▶ Cosine similarity loss

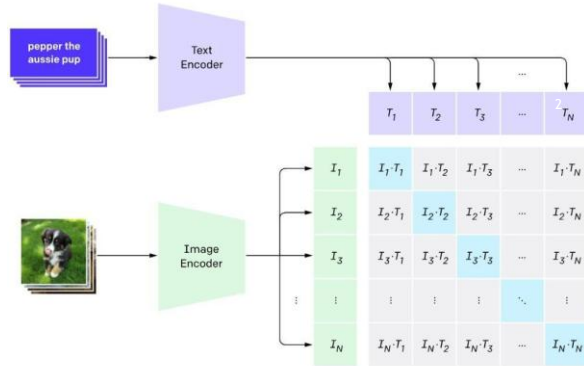
Paper: CLIP: Learning Transferable Visual Models From Natural Language Supervision

<https://arxiv.org/abs/2103.00020>

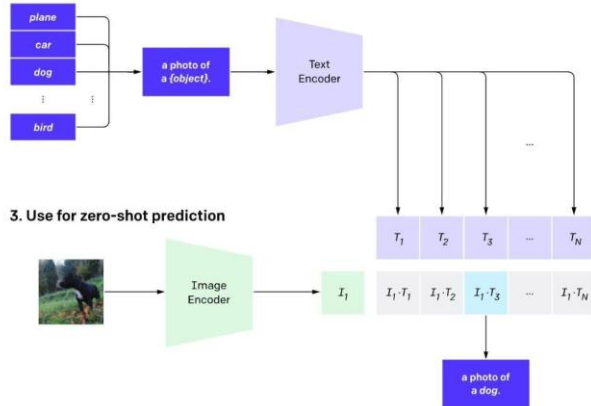
CLIP (Radford et al. 2021)

Exact same contrastive loss as earlier, but.. Transformers and *web data*!

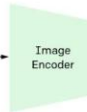
1. Contrastive pre-training



2. Create dataset classifier from label text



3. Use for zero-shot prediction



I_1



a photo of a dog.

CLIP Robustness

Dataset

ImageNet ResNet101 CLIP ViT-L



76.2%

76.2%

ImageNet



64.3%

70.5%

ImageNet V2



37.7%

88.9%

ImageNet Rendition



32.6%

72.3%

ObjectNet



25.2%

60.2%

ImageNet Sketch



2.7%

77.7%

ImageNet Adversarial



- ▶ Joint Transformer-based model: processes both image regions and text.
- ▶ **Early fusion:** image embeddings are added as input tokens to the Transformer.
- ▶ **Pretraining objectives:**
 - Masked Language Modeling (MLM)
 - Next Sentence Prediction (NSP)
 - Image-Text Alignment

Applications:

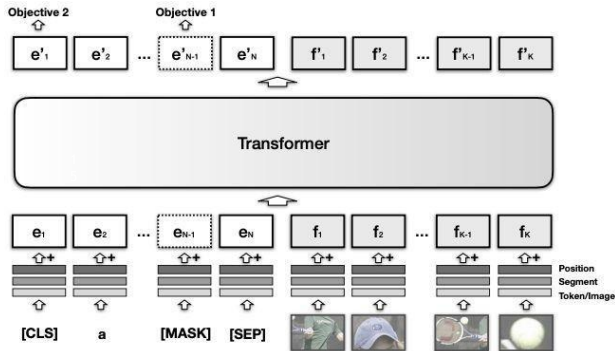
- ▶ Visual Question Answering (VQA)
- ▶ Image captioning
- ▶ Visual commonsense reasoning

Paper: VisualBERT: A Simple and Performant Baseline for Vision and Language

Visual BERTs: VisualBERT

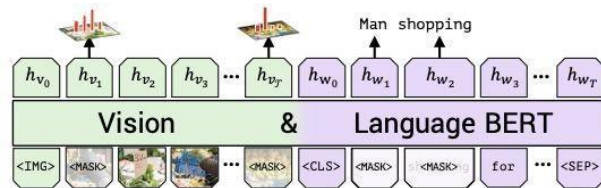
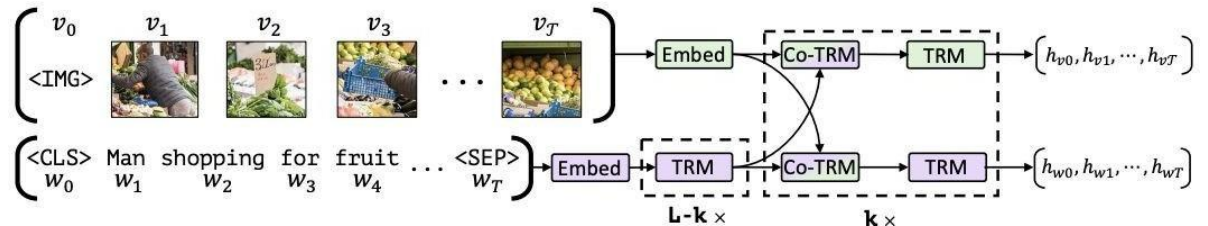


A person hits a ball with a tennis racket

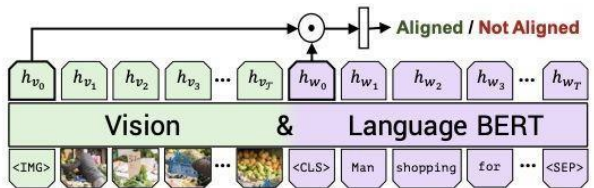


VisualBERT Li et al. 2019

Visual BERTs: ViLBERT



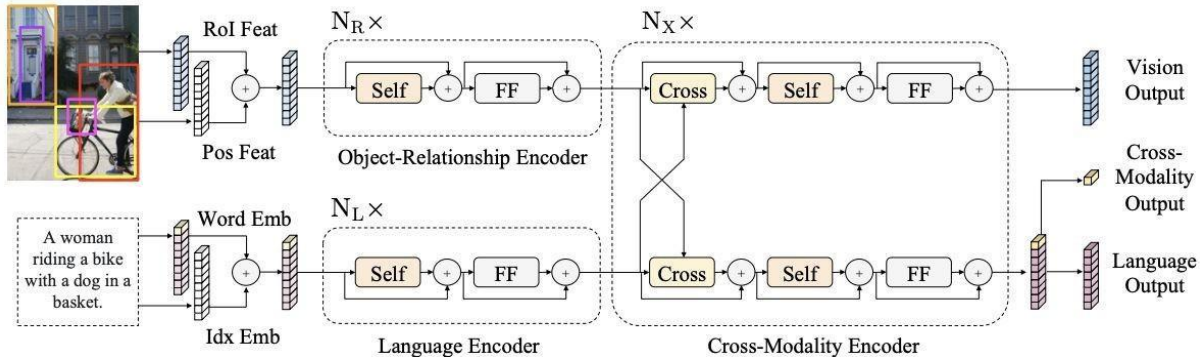
(a) Masked multi-modal learning



(b) Multi-modal alignment prediction

Visual BERTs: LXMERT

Learning Cross-Modality Encoder Representations from Transformers



LXMERT Tan & Bansal 2019

Visual BERTs: Supervised Multimodal Bitransformers

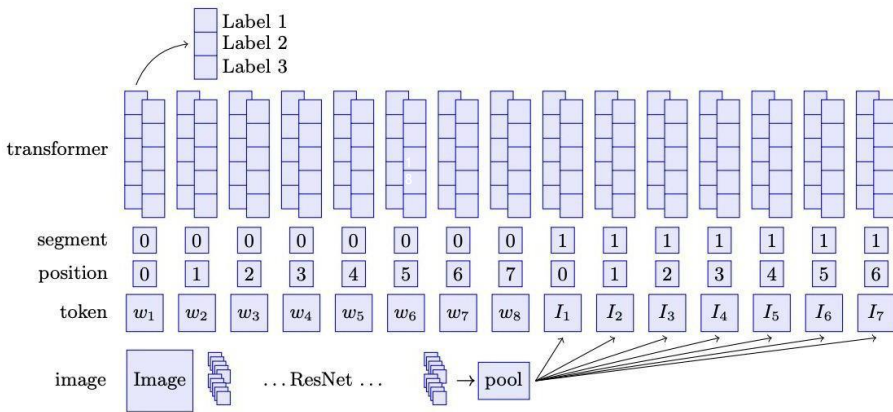
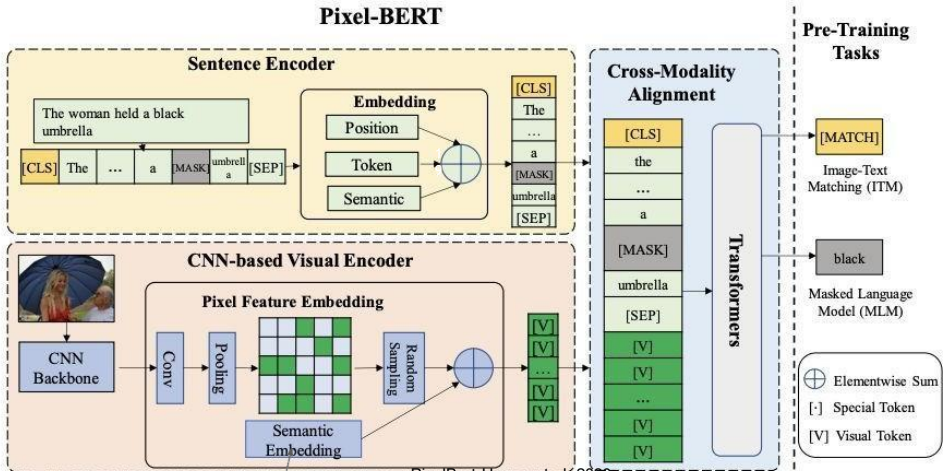


Figure 1: Illustration of the multimodal bitransformer architecture.

Visual BERTs: PixelBert



Misnomer: they mean segment embedding PixelBert Huang et al. 2020

- ▶ FLAVA = **Fusion of Language And Vision Architecture**
- ▶ Handles both unimodal and multimodal tasks
- ▶ **Three towers:** vision, language, multimodal
- ▶ Self-supervised objectives: masked modeling, contrastive learning

Capabilities:

- ▶ Text classification
- ▶ Image classification
- ▶ Visual Question Answering (VQA)
- ▶ Image captioning

Paper: FLAVA: A Foundational Language And Vision Alignment Model

Holistic approach to multimodality.
One foundation model spanning V&L, CV and
NLP. Jointly pre-trained on:

- unimodal text data (CCNews + BookCorpus)
- unimodal image data (ImageNet)
- public paired image-text data (70M)

All data/models are publicly released.



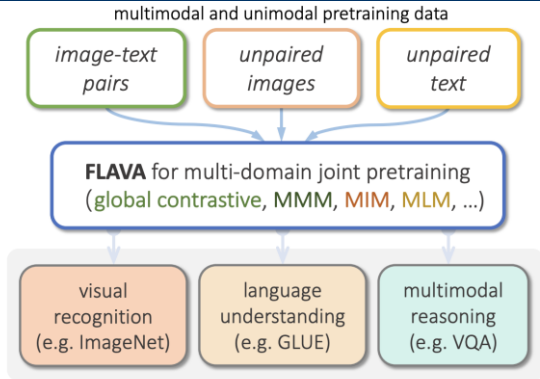


Figure 1. We present FLAVA, a language and vision alignment model that learns strong representations from multimodal (image-text pairs) and unimodal data (unpaired images and text) and can be applied to target a broad scope of tasks from three domains (visual recognition, language understanding, and multimodal reasoning) under a common transformer model architecture.

FLAVA (Singh et al., 2021)

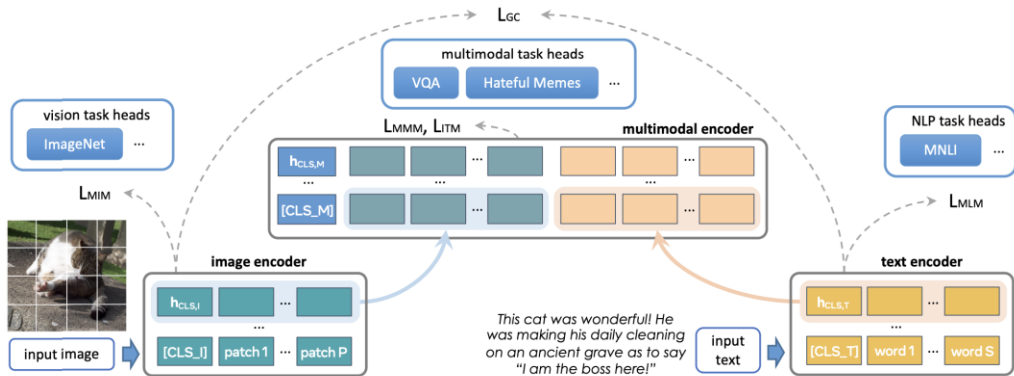


Figure 2. **An overview of our FLAVA model**, with an image encoder transformer to capture unimodal image representations, a text encoder transformer to process unimodal text information, and a multimodal encoder transformer that takes as input the encoded unimodal image and text and integrates their representations for multimodal reasoning. **During pretraining**, masked image modeling (MIM) and mask language modeling (MLM) losses are applied onto the image and text encoders over a single image or a text piece, respectively, while contrastive, masked multimodal modeling (MMM), and image-text matching (ITM) loss are used over paired image-text data. **For downstream tasks**, classification heads are applied on the outputs from the image, text, and multimodal encoders respectively for visual recognition, language understanding, and multimodal reasoning tasks.

LLaVA: Visual Instruction Tuning

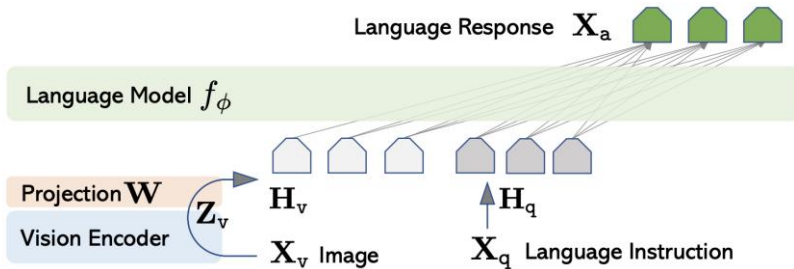
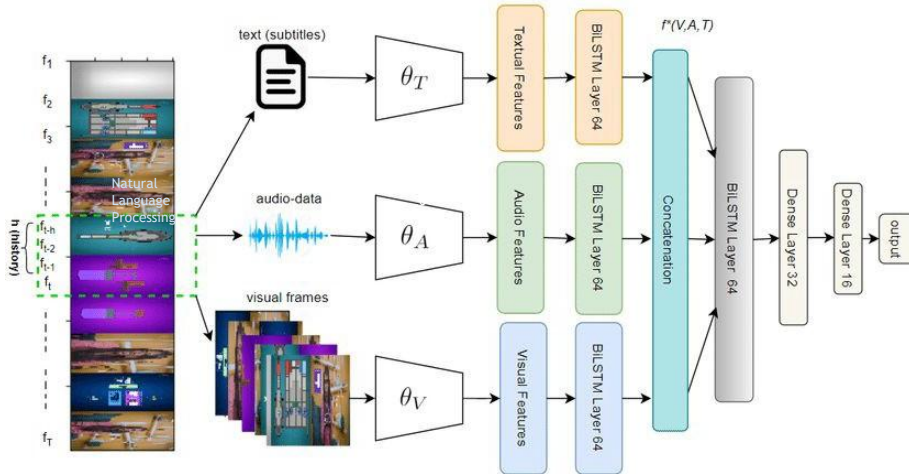


Figure 1: LLaVA network architecture.

Multimodal Architectures



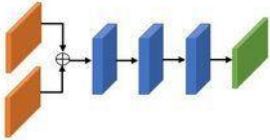
Categories of Multimodal Architectures

- ▶ **Early Fusion:** Combine raw modalities at the input level.
Example: VisualBERT adds image embeddings as tokens to the Transformer.
- ▶ **Late Fusion:** Process each modality separately, then merge representations.
Example: CLIP encodes images and text independently, then compares embeddings.
- ▶ **Hybrid / Cross-modal Attention:** Enable interactions between modalities at multiple stages using attention mechanisms.

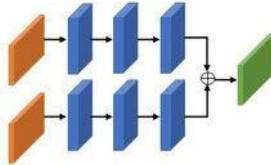
Design Considerations:

- ▶ **Alignment:** How to align representations across modalities.
- ▶ **Modality Dropout:** Handling missing or noisy modalities during training or inference.
- ▶ **Shared vs. Modality-Specific Layers:** Deciding which layers are shared and which are unique to each modality.

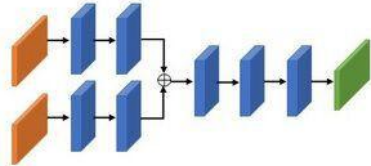
Categories of Multimodal Architectures



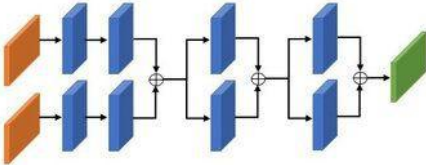
(a) Early Fusion



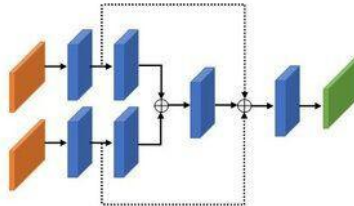
(b) Late Fusion



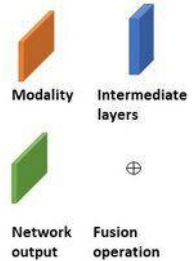
(c) Middle Fusion - fusion in one layer



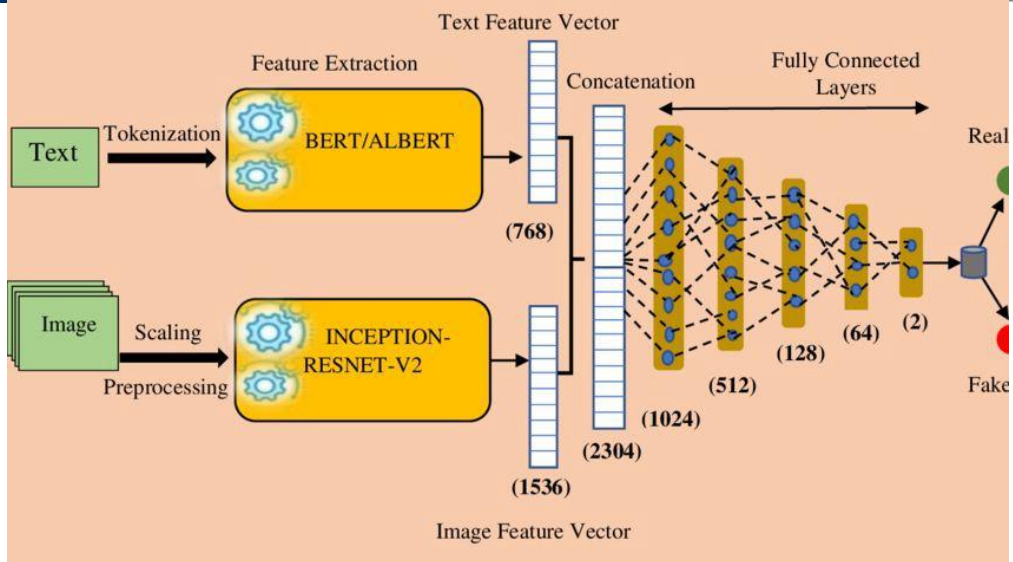
(d) Middle Fusion - deep fusion



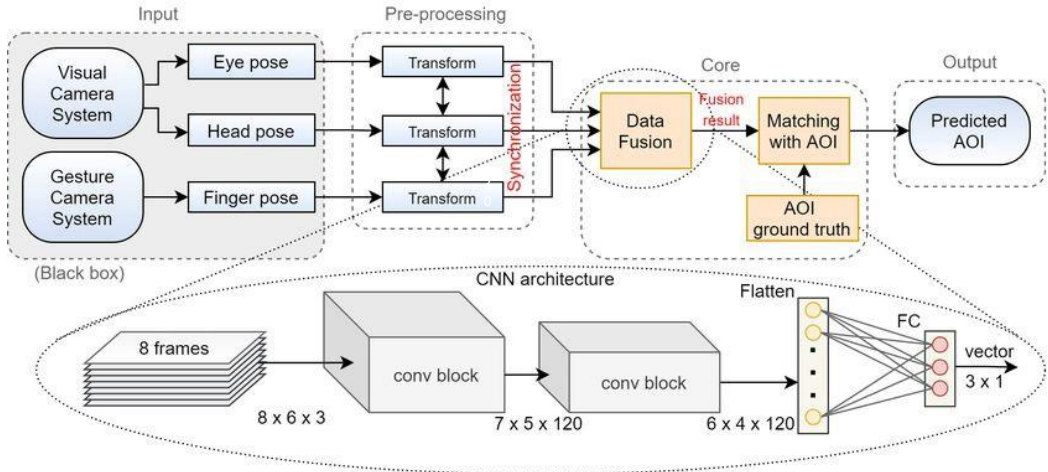
(e) Middle Fusion - short-cut fusion



Early Fusion



Late Fusion



Challenges in Multimodal Integration

MultiModal Learning Challenges

Multimodal Machine Learning Challenges

Representation

Finding a good representation is crucial for the success of multimodal models that combine different modalities such as image, text, and audio.

Fusion

Fusing information from different modalities to perform a prediction task is the core of multimodal learning and can be challenging due to the diverse nature of multimodal data.

Alignment

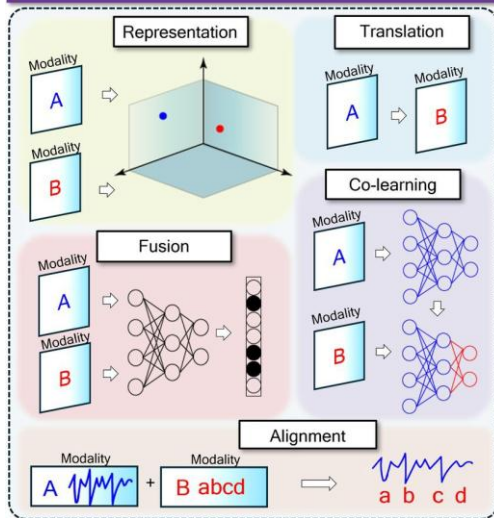
Aligning data from different modalities to create modality-invariant representations is necessary to combine information effectively.

Translation

Translating data from one modality to another is a crucial task in multimodal learning, but can be difficult due to the subjective and open-ended nature of translations.

Co-learning

Co-learning aims to transfer knowledge learned through one or more modalities to tasks involving another. It is useful in medical diagnosis combining CT and MRI scans.

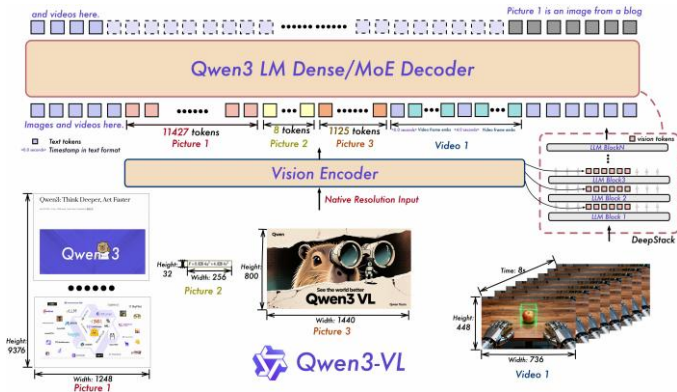


Multimodal LLMs in 2025–2026

- **Qwen2.5-VL** (Feb 2025): a dynamic-resolution ViT trained from scratch with window attention; absolute time encoding gives hours-long video comprehension with second-level event localization; grounding by bounding box or point; structured parsing of invoices, forms and tables.
- **Qwen3-VL** (Sep–Nov 2025): dense and MoE sizes from 2B to 235B, each in Instruct and reasoning-focused Thinking editions; native 256K context, expandable to 1M; operates PC and mobile GUIs as a visual agent; OCR in 32 languages.
- **InternVL3 / InternVL3.5** (2025): *native multimodal pre-training*, one joint stage over text and multimodal corpora instead of retrofitting a finished text LLM; InternVL3.5 adds Cascade RL and a Visual Resolution Router for a $4\times$ inference speedup, reporting state of the art among open-source MLLMs.

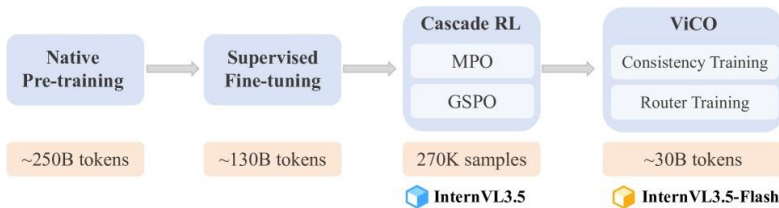
Bai et al., [arXiv:2502.13923](https://arxiv.org/abs/2502.13923); Bai et al., [arXiv:2511.21631](https://arxiv.org/abs/2511.21631); Zhu et al., [arXiv:2504.10479](https://arxiv.org/abs/2504.10479); Wang et al., [arXiv:2508.18265](https://arxiv.org/abs/2508.18265).

Inside a 2025 Flagship: Qwen3-VL



Native-resolution images and timestamped video frames enter one decoder as interleaved tokens, so token count scales with input size: 11,427 tokens for a long webpage, 8 for a small formula. DeepStack feeds multi-level ViT features into several LLM blocks. Source: Qwen team, Qwen3-VL repository, [arXiv:2511.21631](https://arxiv.org/abs/2511.21631).

- **Encoder first:** Qwen2.5-VL trains its ViT alone on 1.5T tokens of captions, OCR and visual knowledge while the LLM stays frozen, then trains ViT and LLM jointly on 2T mixed multimodal tokens.
- **Then supervised fine-tuning** on multimodal instruction data, exactly as for text LLMs.
- **Then reinforcement learning:** InternVL3.5's Cascade RL runs an offline preference stage (MPO) and then an online stage (GSPO). RL post-training algorithms are covered later in the course.



InternVL3.5's recipe: native pre-training, supervised fine-tuning, Cascade RL; the final ViCO stage trains the Flash variant's resolution router. Source: Wang et al., [arXiv:2508.18265](https://arxiv.org/abs/2508.18265), Figure 3.

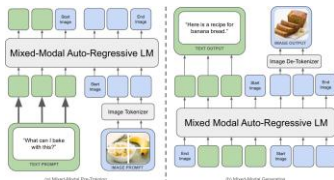
- **GPT-4o** (May 2024): “a single new model end-to-end across text, vision, and audio”; the same network reads and produces all of them.
- **Gemini 2.5 Pro** (Mar 2025): text, audio, images, video and whole code repositories in a 1M-token context window.
- **GPT-5** (Aug 2025): one unified system that decides when to think longer; 84.2% on MMMU at launch.
- **Gemini 3** (Nov 2025): launched as “the best model in the world for multimodal understanding”: 81% on MMMU-Pro, 87.6% on Video-MMMU.
- **2026 so far**: new releases land every few months (GPT-5.4 through 5.6, Gemini 3.5 and 3.6, Claude Opus 4.7 and Fable 5); the best frontier MMMU-Pro score moved from 81% to 83% without tools (84.6% with tools) in eight months.

Sources: OpenAI, Google and Anthropic release announcements, 2024–2026 (see references).

- **OpenAI o3 and o4-mini** (Apr 2025) were the first OpenAI models that “think with images in their chain-of-thought” rather than merely looking at them.
- During reasoning the model crops, zooms into and rotates the user’s image, natively and without separate specialist models, and can mix these steps with web search and Python.
- Result: state of the art at the time on MMMU, MathVista and chart reasoning, and 95.7% on the V* visual-search benchmark, “largely solving” it.
- The pattern generalizes: perception becomes a sequence of *actions* on the input, interleaved with textual reasoning. How such reasoning is trained is covered later in the course.

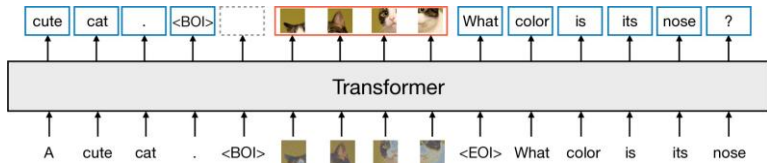
OpenAI, “Thinking with images”, April 2025, openai.com/index/thinking-with-images.

- The projector recipe bolts vision onto a finished LLM. The native alternative pretrains one transformer on every modality tokenized together: **Chameleon** (Meta, 2024) maps a 512×512 image to 1024 discrete tokens, and one autoregressive model reads and writes both modalities.
- **Llama 4** (Apr 2025): Meta's first open-weight family pretrained with early fusion, jointly on text, image and video data.
- Across 457 trained models, late fusion showed no inherent advantage; early fusion is stronger at small scales and cheaper to train (Shukor et al., ICCV 2025).



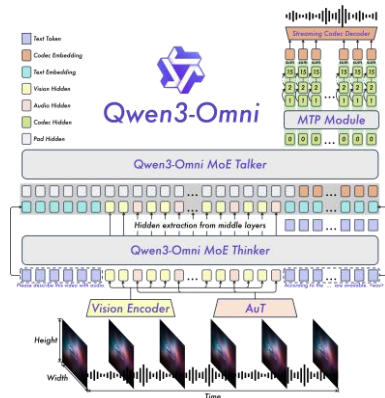
Source: Chameleon Team, [arXiv:2405.09818](https://arxiv.org/abs/2405.09818), Figure 1.

- **Transfusion** (ICLR 2025): one transformer, two losses: next-token prediction on text tokens, a diffusion objective on continuous image patches in the same sequence; scales better than quantizing images.
- **Janus-Pro** (DeepSeek, Jan 2025): decoupled visual encoding: SigLIP features for understanding, VQ tokens for generation, one shared autoregressive core.
- **In production:** GPT-4o image generation (Mar 2025) and Gemini 2.5 Flash Image (“Nano Banana”, Aug 2025) put native image generation and editing in assistants; Nano Banana Pro (Nov 2025) grounds generations in live search.



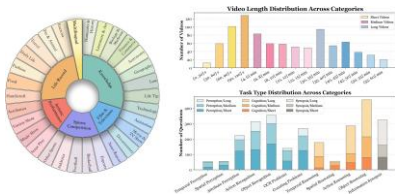
One sequence mixes text tokens with continuous image patches between `<BOI>`/`<EOI>`. Source: Zhou et al., ICLR 2025, [arXiv:2408.11039](https://arxiv.org/abs/2408.11039), Figure 1.

- **GPT-4o** answers speech in 320 ms on average; the cascaded pipeline it replaced (transcription, text LLM, TTS) took 2.8–5.4 s and lost tone, multiple speakers and laughter.
- **Qwen2.5-Omni** (Mar 2025): **Thinker-Talker**: the Thinker LLM writes text while the Talker streams speech tokens from its hidden states; TMRoPE keeps audio and video frames on one timeline.
- **Qwen3-Omni** (Sep 2025): both parts become MoE; 234 ms theoretical first-packet latency; open-source state of the art on 32 of 36 audio and audio-visual benchmarks. **Qwen3.5-Omni** (Apr 2026) scales to a 256K context and over ten hours of audio input.
- Speech and audio processing get their own treatment later in the course.



Source: Xu et al., “Qwen3-Omni Technical Report”, [arXiv:2509.17765](https://arxiv.org/abs/2509.17765), Figure 2.

- Video enters the model as resized frames plus timestamps. Gemini spends roughly 300 tokens per second of video, so a 1M-token context holds about an hour.
- Qwen2.5-VL's absolute time encoding localizes events to the second inside hours-long video.
- **Video-MME** (CVPR 2025): 900 videos totalling 254 hours with 2,700 expert QA pairs, durations from 11 seconds to an hour. Gemini 1.5 Pro led at release with 75% (no subtitles), and every model degrades as the video gets longer.

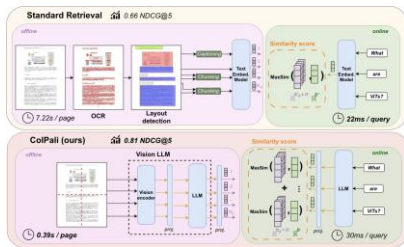


Source: Fu et al., "Video-MME", CVPR 2025, [arXiv:2405.21075](https://arxiv.org/abs/2405.21075), Figure 2.

- Invoices, forms, tables, charts and scanned pages are where most users meet these models. DocVQA and ChartQA (covered earlier) defined the tasks; document parsing is now a headline capability of every flagship.
- **Qwen2.5-VL** turns document pages into structured output and analyzes charts, diagrams and layouts.
- **OmniDocBench** (CVPR 2025) evaluates end-to-end document parsing across nine source types, from academic papers and textbooks to handwriting and densely typeset newspapers.
- Frontier labs report against it: OpenAI cites GPT-5.4 cutting its OmniDocBench error to 0.109 from GPT-5.2's 0.140.

Ouyang et al., [arXiv:2412.07626](https://arxiv.org/abs/2412.07626); OpenAI, "Introducing GPT-5.4", March 2026, openai.com/index/introducing-gpt-5-4.

- Retrieval-augmented generation (covered earlier) assumes clean text. Real corpora arrive as page images whose figures, tables and layout an OCR pipeline flattens or loses.
- ColPali** (ICLR 2025) embeds the page image directly: a vision-language model produces multi-vector embeddings, matched to the query with ColBERT-style late interaction.
- It outperforms the OCR pipelines while indexing pages an order of magnitude faster: 0.39 s against 7.22 s per page in the paper's comparison.

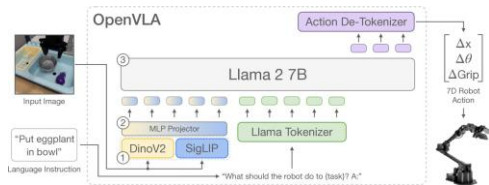


Source: Faysse et al., "ColPali", ICLR 2025, [arXiv:2407.01449](https://arxiv.org/abs/2407.01449), Figure 1.

- **The perception task:** map an instruction to the right pixel. ScreenSpot (ACL 2024) tests grounding on 600+ screenshots across mobile, desktop and web; ScreenSpot-Pro moves to high-resolution professional software, where the best grounding model at its release managed 18.9%.
- **The benchmark:** OSWorld (NeurIPS 2024): 369 real tasks in real operating systems, scored by executing the resulting state.
- **The products:** Claude computer use (Oct 2024, the first frontier model to offer it), OpenAI Operator (Jan 2025) and Gemini 2.5 Computer Use (Oct 2025) all run the same loop: screenshot in, click or keystroke out.
- **The trajectory:** Anthropic reports its models moving from under 15% on OSWorld in late 2024 to 72.5% by early 2026. The agent architectures around this loop are covered later in the course.

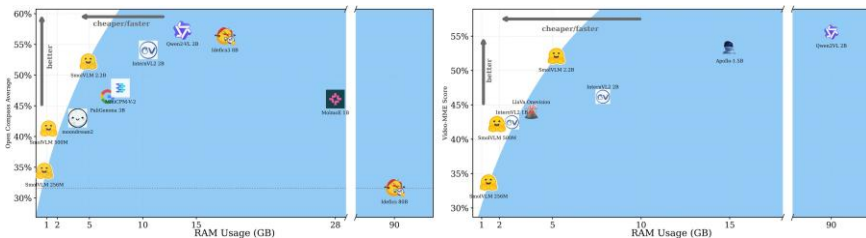
Cheng et al., [arXiv:2401.10935](https://arxiv.org/abs/2401.10935); Li et al., [arXiv:2504.07981](https://arxiv.org/abs/2504.07981); Xie et al., [arXiv:2404.07972](https://arxiv.org/abs/2404.07972); Anthropic, Feb 2026, anthropic.com/news/acquires-vercept.

- **RT-2** (CoRL 2023): fine-tune a VLM on robot trajectories with actions written as text tokens; web-scale knowledge transfers to manipulation.
- **OpenVLA** (CoRL 2024): open 7B model trained on 970k demonstrations; fused DINOv2 and SigLIP features into a Llama 2 backbone, 7-D actions out.
- π_0 (2024): a 300M flow-matching action expert on a PaliGemma VLM emits continuous action chunks; $\pi_{0.5}$ (2025) generalizes to unseen homes.
- **Gemini Robotics** (2025): Gemini 2.0 as a VLA; the 1.5 release pairs an embodied-reasoning VLM with a separate acting VLA it instructs.



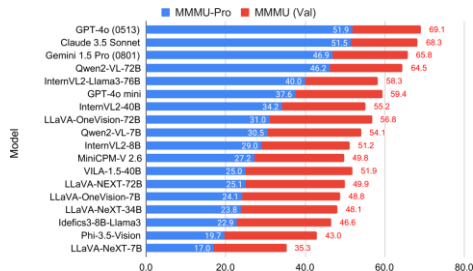
Source: Kim et al., "OpenVLA", CoRL 2024, [arXiv:2406.09246](https://arxiv.org/abs/2406.09246), Figure 1.

- **SmoIVLM** (2025): 256M, 500M and 2.2B models; the smallest runs in under 1 GB of GPU memory.
- **PaliGemma 2** (2024) and **Gemma 3** (2025): SigLIP encoders on Gemma LLMs; Gemma 3's 4B, 12B and 27B models share one frozen 417M-parameter encoder.
- **Most visual tokens are wasted anyway:** FastV (ECCV 2024) prunes half of them after layer 2, training-free, cutting FLOPs by 45% with negligible loss.



Benchmark average against GPU memory. Source: Marafioti et al., "SmoIVLM", [arXiv:2504.05299](https://arxiv.org/abs/2504.05299), Figure 1.

- **MMMU** (CVPR 2024): 11.5K college-level questions across six disciplines, 30 subjects and 30 image types, from charts and maps to music sheets and chemical structures. GPT-4V managed 56% at launch; the strongest human-expert baseline is 88.6%.
- **MMMU-Pro** (ACL 2025) filters out questions text-only models can answer, widens choices to ten, and adds a vision-only setting where the question itself is an image. Scores drop by 16.8 to 26.9 points.
- By mid-2026 the best systems reach the mid-80s on MMMU and the low-80s on MMMU-Pro, brushing the estimated 85.4% human-expert baseline there.

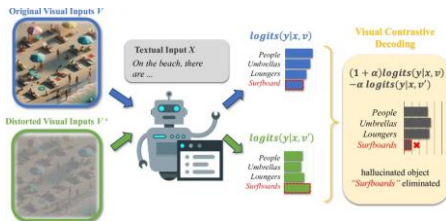


The same models on both benchmarks: every model loses ground on MMMU-Pro. Source: Yue et al., "MMMU-Pro", ACL 2025, [arXiv:2409.02813](https://arxiv.org/abs/2409.02813) (v1), Figure 1.

- **Object hallucination.** Multimodal LLMs describe objects that are not in the image, especially objects that are frequent in, or co-occur with, what they saw in training. POPE (EMNLP 2023) measures this with polling-style yes/no probes.
- **Visual tokens are expensive.** A long webpage screenshot costs 11,427 tokens in Qwen3-VL's own example. DeepSeek-OCR (Oct 2025) inverts the problem, rendering text as an image and reading it back: at a compression ratio below $10\times$ it decodes with 97% precision ("contexts optical compression").
- **Evaluation is fragile.** Benchmark questions leak into training data, and multiple-choice scores overstate perception: MMMU-Pro's vision-only setting exists because models were often reading the question, not the image.
- **The attack surface grows with each modality:** instructions can be embedded in images and audio where text-level filters never look. Safety for agentic systems is covered later in the course.

Li et al., [arXiv:2305.10355](https://arxiv.org/abs/2305.10355); Wei et al., [arXiv:2510.18234](https://arxiv.org/abs/2510.18234).

- **Fix the data:** RLHF-V (CVPR 2024) collects segment-level human *corrections* of hallucinated spans and trains on them with dense preference optimization; 1.4k fine-grained corrections already cut object hallucination sharply.
- **Fix the decoding:** Visual Contrastive Decoding (CVPR 2024) needs no training at all: contrast the token distribution given the real image against the one given a noised copy, and the language prior's contribution cancels out.



Source: Leng et al., “Visual Contrastive Decoding”, CVPR 2024, [arXiv:2311.16922](https://arxiv.org/abs/2311.16922), Figure 1.

- **A single image can break alignment:** Qi et al. (AAAI 2024) optimize one adversarial image that universally jailbreaks an aligned vision-language model, eliciting harmful behavior far beyond the small corpus used to build the image.
- **Injection hides in pixels and audio:** Bagdasaryan et al. (2023) blend an instruction-carrying perturbation into an image or sound clip; when a user innocently asks about the file, the model follows the hidden instruction.
- **Why it works:** safety training lives largely in text space, while the continuous, high-dimensional visual channel routes around it.
- Defenses lag the attacks; a deployed multimodal system has to treat every pixel as untrusted input.

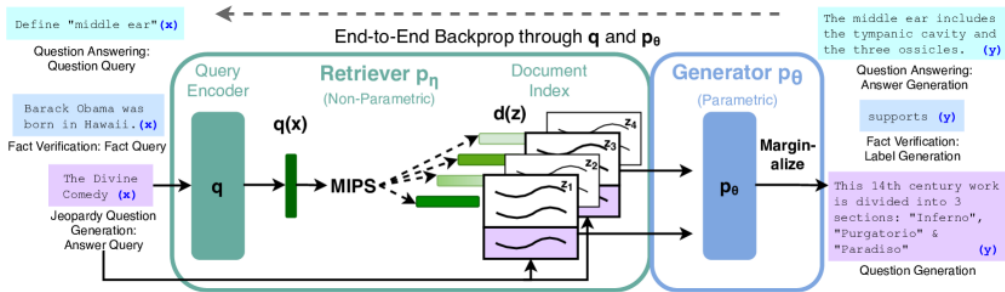
Qi et al., [arXiv:2306.13213](https://arxiv.org/abs/2306.13213); Bagdasaryan et al., [arXiv:2307.10490](https://arxiv.org/abs/2307.10490).

Retrieval-Augmented Generation (RAG)

- ▶ LLMs are fixed after training: their knowledge cannot be easily updated.
- ▶ Updating facts or adding new information requires retraining or fine-tuning, which is costly and slow.
- ▶ Retrieval-Augmented Generation (RAG) addresses this by combining:
 - A **retriever** that fetches relevant documents or facts from an external knowledge base.
 - A **reader** (LLM) that generates answers conditioned on the retrieved information.
- ▶ This enables up-to-date, accurate, and context-aware responses without retraining the LLM.

Two main components:

- **Retriever:** Encodes the query and the documents into vectors and fetches the closest passages from the corpus by similarity search (e.g., DPR, ColBERT).
- **Generator** (also called the **reader**): An LLM that produces the final answer conditioned on the retrieved passages (e.g., BART, GPT).



Source: Lewis et al., NeurIPS 2020, Fig. 1.

- 1. User asks a question.**
- 2. Retriever:** Embedding and similarity search retrieves top- k relevant documents from the knowledge base.
- 3. Reader:** Retrieved documents and the user prompt are fed to the LLM.
- 4. LLM generates a response** grounded in the retrieved documents.

- ▶ **LangChain:** Framework for building LLM-powered applications with retrieval, chaining, and orchestration.
- ▶ **LlamaIndex:** Data framework for connecting LLMs with external data sources and knowledge bases.
- ▶ **Haystack:** Open-source framework for building production-ready RAG pipelines.

Popular Vector Databases:

- ▶ **FAISS:** Facebook AI Similarity Search, efficient vector search library.
- ▶ **Weaviate:** Scalable, cloud-native vector database with RESTful APIs.
- ▶ **Qdrant:** Open-source vector search engine with filtering and payload support.

- ▶ **Factual accuracy:** Answers are grounded in retrieved documents, reducing hallucinations.
- ▶ **Real-time updates:** Knowledge can be updated instantly by modifying the underlying corpus, without retraining the LLM.
- ▶ **Scalable with large corpora:** Efficient retrieval enables handling vast and dynamic knowledge bases.

- ▶ **Requires good retrieval quality:** Poor retrieval leads to irrelevant or incomplete context for the LLM.
- ▶ **Latency overhead:** Retrieval and reranking steps add to response time.
- ▶ **Domain-specific retrievers may require tuning:** Customization and fine-tuning are often needed for specialized domains.
- ▶ **Context length limits in LLMs:** Only a limited amount of retrieved content can be passed to the LLM at once.

- ▶ **Contrastive alignment** (CLIP, then SigLIP) gives encoders that score and retrieve; conversation took a decoder LLM behind a projector plus visual instruction tuning.
- ▶ **The open recipe scaled** through LLaVA's successors into today's flagships (Qwen3-VL, InternVL3.5): native resolution, grounding, documents, video.
- ▶ **Frontier systems are natively multimodal**: one network reads and writes text, images and audio (GPT-4o onward), reasons by acting on images (o3), and generates images mid-conversation (Nano Banana).
- ▶ **The same perception stack works beyond chat**: retrieving page images, driving GUIs, controlling robots.
- ▶ **Where fusion happens** (early, middle, late) is still the design axis; at scale, early fusion concedes nothing.
- ▶ **Benchmarks brush the human-expert ceiling** (MMMU-Pro in the low-80s); hallucination, token cost, jailbreaks and fragile evaluation stay open.
- ▶ The agentic systems built on these models, their loops, tools and safety questions, are covered later in the course.

- 1 Kiela, D. (2023).
Multimodal Deep Learning (with an NLP focus). Guest lecture, Stanford CS224n, Winter 2023.
[Slides \(PDF\)](#)
- 2 McGurk, H., MacDonald, J. (1976).
Hearing lips and seeing voices.
Nature 264(5588):746–748. [doi:10.1038/264746a0](https://doi.org/10.1038/264746a0)
- 3 Harnad, S. (1990).
The Symbol Grounding Problem.
Physica D 42:335–346. [arXiv:cs/9906002](https://arxiv.org/abs/cs/9906002)
- 4 Bender, E. M., Koller, A. (2020).
Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data.
ACL 2020, pp. 5185–5198. aclanthology.org/2020.acl-main.463
- 5 Vinyals, O., Toshev, A., Bengio, S., Erhan, D. (2015).
Show and Tell: A Neural Image Caption Generator.
CVPR 2015. [arXiv:1411.4555](https://arxiv.org/abs/1411.4555)

- 6 Antol, S., Agrawal, A., Lu, J., et al. (2015).
VQA: Visual Question Answering.
ICCV 2015. [arXiv:1505.00468](https://arxiv.org/abs/1505.00468)
- 7 Goyal, Y., et al. (2017).
Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering.
CVPR 2017. [arXiv:1612.00837](https://arxiv.org/abs/1612.00837)
- 8 Chen, X., et al. (2015).
Microsoft COCO Captions: Data Collection and Evaluation Server.
[arXiv:1504.00325](https://arxiv.org/abs/1504.00325)
- 9 Yu, L., et al. (2016).
Modeling Context in Referring Expressions.
ECCV 2016. [arXiv:1608.00272](https://arxiv.org/abs/1608.00272)
- 10 Singh, A., et al. (2019).
Towards VQA Models That Can Read.
CVPR 2019. [arXiv:1904.08920](https://arxiv.org/abs/1904.08920)
- 11 Mathew, M., et al. (2021).
DocVQA: A Dataset for VQA on Document Images.
WACV 2021. [arXiv:2007.00398](https://arxiv.org/abs/2007.00398)

- 12 Masry, A., et al. (2022).
ChartQA: A Benchmark for Question Answering about Charts with Visual and Logical Reasoning.
Findings of ACL 2022. [arXiv:2203.10244](https://arxiv.org/abs/2203.10244)
- 13 Radford, A., Kim, J. W., Hallacy, C., et al. (2021).
Learning Transferable Visual Models From Natural Language Supervision.
ICML 2021, PMLR 139:8748–8763. [arXiv:2103.00020](https://arxiv.org/abs/2103.00020)
- 14 Zhai, X., Mustafa, B., Kolesnikov, A., Beyer, L. (2023).
Sigmoid Loss for Language Image Pre-Training.
ICCV 2023. [arXiv:2303.15343](https://arxiv.org/abs/2303.15343)
- 15 Tschannen, M., Gritsenko, A., Wang, X., et al. (2025).
SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features.
[arXiv:2502.14786](https://arxiv.org/abs/2502.14786)
- 16 Li, L. H., Yatskar, M., Yin, D., Hsieh, C. J., Chang, K. W. (2019).
VisualBERT: A Simple and Performant Baseline for Vision and Language.
[arXiv:1908.03557](https://arxiv.org/abs/1908.03557)

- 17 Lu, J., Batra, D., Parikh, D., Lee, S. (2019).
ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks.
NeurIPS 2019. [arXiv:1908.02265](https://arxiv.org/abs/1908.02265)
- 18 Tan, H., Bansal, M. (2019).
LXMERT: Learning Cross-Modality Encoder Representations from Transformers.
EMNLP 2019. [arXiv:1908.07490](https://arxiv.org/abs/1908.07490)
- 19 Huang, Z., Zeng, Z., Liu, B., Fu, D., Fu, J. (2020).
Pixel-BERT: Aligning Image Pixels with Text by Deep Multi-Modal Transformers.
[arXiv:2004.00849](https://arxiv.org/abs/2004.00849)
- 20 Singh, A., Hu, R., Goswami, V., et al. (2021).
FLAVA: A Foundational Language and Vision Alignment Model.
CVPR 2022. [arXiv:2112.04482](https://arxiv.org/abs/2112.04482)
- 21 Alayrac, J.-B., Donahue, J., Luc, P., et al. (2022).
Flamingo: a Visual Language Model for Few-Shot Learning.
NeurIPS 2022. [arXiv:2204.14198](https://arxiv.org/abs/2204.14198)

- 22 Li, J., Li, D., Savarese, S., Hoi, S. (2023).
BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models.
ICML 2023, PMLR 202:19730–19742. [arXiv:2301.12597](https://arxiv.org/abs/2301.12597)
- 23 Liu, H., Li, C., Wu, Q., Lee, Y. J. (2023).
Visual Instruction Tuning (LLaVA).
NeurIPS 2023 (Oral). [arXiv:2304.08485](https://arxiv.org/abs/2304.08485)
- 24 Liu, H., Li, C., Li, Y., Lee, Y. J. (2024).
Improved Baselines with Visual Instruction Tuning (LLaVA-1.5).
CVPR 2024. [arXiv:2310.03744](https://arxiv.org/abs/2310.03744)
- 25 Li, B., Zhang, Y., et al. (2024).
LLaVA-OneVision: Easy Visual Task Transfer.
[arXiv:2408.03326](https://arxiv.org/abs/2408.03326)
- 26 Deitke, M., et al. (2024).
Molmo and PixMo: Open Weights and Open Data for State-of-the-Art Vision-Language Models.
CVPR 2025. [arXiv:2409.17146](https://arxiv.org/abs/2409.17146)

- 27 Bai, S., Chen, K., Liu, X., et al. (2025).
Qwen2.5-VL Technical Report.
[arXiv:2502.13923](https://arxiv.org/abs/2502.13923)
- 28 Bai, S., Cai, Y., Chen, R., et al. (2025).
Qwen3-VL Technical Report.
[arXiv:2511.21631](https://arxiv.org/abs/2511.21631)
- 29 Zhu, J., et al. (2025).
InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models.
[arXiv:2504.10479](https://arxiv.org/abs/2504.10479)
- 30 Wang, W., Gao, Z., Gu, L., et al. (2025).
InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency.
[arXiv:2508.18265](https://arxiv.org/abs/2508.18265)
- 31 Wang, W., et al. (2024).
Enhancing the Reasoning Ability of Multimodal Large Language Models via Mixed Preference Optimization.
[arXiv:2411.10442](https://arxiv.org/abs/2411.10442)

- 32 Chameleon Team (2024).
Chameleon: Mixed-Modal Early-Fusion Foundation Models.
[arXiv:2405.09818](https://arxiv.org/abs/2405.09818)
- 33 Zhou, C., et al. (2024).
Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model.
ICLR 2025. [arXiv:2408.11039](https://arxiv.org/abs/2408.11039)
- 34 Chen, X., et al. (2025).
Janus-Pro: Unified Multimodal Understanding and Generation with Data and Model Scaling.
[arXiv:2501.17811](https://arxiv.org/abs/2501.17811)
- 35 Shukor, M., Fini, E., et al. (2025).
Scaling Laws for Native Multimodal Models.
ICCV 2025. [arXiv:2504.07951](https://arxiv.org/abs/2504.07951)
- 36 Meta AI (2025).
The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation.
ai.meta.com/blog/llama-4-multimodal-intelligence
- 37 Xu, J., et al. (2025).
Qwen2.5-Omni Technical Report.
[arXiv:2503.20215](https://arxiv.org/abs/2503.20215)

- 38 Xu, J., et al. (2025).
Qwen3-Omni Technical Report.
[arXiv:2509.17765](https://arxiv.org/abs/2509.17765)
- 39 Qwen Team (2026).
Qwen3.5-Omni Technical Report.
[arXiv:2604.15804](https://arxiv.org/abs/2604.15804)
- 40 Fu, C., et al. (2025).
Video-MME: The First-Ever Comprehensive Evaluation Benchmark of Multi-modal LLMs in Video Analysis.
CVPR 2025. [arXiv:2405.21075](https://arxiv.org/abs/2405.21075)
- 41 Ouyang, L., et al. (2025).
OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations.
CVPR 2025. [arXiv:2412.07626](https://arxiv.org/abs/2412.07626)
- 42 Faysse, M., et al. (2025).
ColPali: Efficient Document Retrieval with Vision Language Models.
ICLR 2025. [arXiv:2407.01449](https://arxiv.org/abs/2407.01449)
- 43 Cheng, K., et al. (2024).
SeeClick: Harnessing GUI Grounding for Advanced Visual GUI Agents.
ACL 2024. [arXiv:2401.10935](https://arxiv.org/abs/2401.10935)

- 44 Li, K., et al. (2025).
ScreenSpot-Pro: GUI Grounding for Professional High-Resolution Computer Use.
ACM Multimedia 2025. [arXiv:2504.07981](https://arxiv.org/abs/2504.07981)

- 45 Xie, T., et al. (2024).
OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments.
NeurIPS 2024 Datasets and Benchmarks. [arXiv:2404.07972](https://arxiv.org/abs/2404.07972)

- 46 Brohan, A., et al. (2023).
RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control.
CoRL 2023. [arXiv:2307.15818](https://arxiv.org/abs/2307.15818)

- 47 Kim, M. J., et al. (2024).
OpenVLA: An Open-Source Vision-Language-Action Model.
CoRL 2024. [arXiv:2406.09246](https://arxiv.org/abs/2406.09246)

- 48 Black, K., et al. (2024).
 π_0 : A Vision-Language-Action Flow Model for General Robot Control.
RSS 2025. [arXiv:2410.24164](https://arxiv.org/abs/2410.24164)

- 49 Gemini Robotics Team, Google DeepMind (2025).
Gemini Robotics: Bringing AI into the Physical World.
[arXiv:2503.20020](https://arxiv.org/abs/2503.20020)

- 50 Marafioti, A., et al. (2025).
SmolVLM: Redefining small and efficient multimodal models.
[arXiv:2504.05299](https://arxiv.org/abs/2504.05299)
- 51 Steiner, A., et al. (2024).
PaliGemma 2: A Family of Versatile VLMs for Transfer.
[arXiv:2412.03555](https://arxiv.org/abs/2412.03555)
- 52 Gemma Team, Google DeepMind (2025).
Gemma 3 Technical Report.
[arXiv:2503.19786](https://arxiv.org/abs/2503.19786)
- 53 Chen, L., et al. (2024).
An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models (FastV).
ECCV 2024. [arXiv:2403.06764](https://arxiv.org/abs/2403.06764)
- 54 Yue, X., Ni, Y., Zhang, K., et al. (2024).
MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI.
CVPR 2024. [arXiv:2311.16502](https://arxiv.org/abs/2311.16502)

- 55 Yue, X., Zheng, T., Ni, Y., et al. (2025).
MMM-Pro: A More Robust Multi-discipline Multimodal Understanding Benchmark.
ACL 2025. [arXiv:2409.02813](https://arxiv.org/abs/2409.02813)
- 56 Li, Y., Du, Y., Zhou, K., Wang, J., et al. (2023).
Evaluating Object Hallucination in Large Vision-Language Models (POPE).
EMNLP 2023. [arXiv:2305.10355](https://arxiv.org/abs/2305.10355)
- 57 Wei, H., Sun, Y., Li, Y. (2025).
DeepSeek-OCR: Contexts Optical Compression.
[arXiv:2510.18234](https://arxiv.org/abs/2510.18234)
- 58 Yu, T., et al. (2024).
RLHF-V: Towards Trustworthy MLLMs via Behavior Alignment from Fine-grained Correctional Human Feedback.
CVPR 2024. [arXiv:2312.00849](https://arxiv.org/abs/2312.00849)
- 59 Leng, S., et al. (2024).
Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding.
CVPR 2024. [arXiv:2311.16922](https://arxiv.org/abs/2311.16922)

- 60 Qi, X., et al. (2024).
Visual Adversarial Examples Jailbreak Aligned Large Language Models.
AAAI 2024. [arXiv:2306.13213](https://arxiv.org/abs/2306.13213)
- 61 Bagdasaryan, E., Hsieh, T.-Y., Nassi, B., Shmatikov, V. (2023).
Abusing Images and Sounds for Indirect Instruction Injection in Multi-Modal LLMs.
[arXiv:2307.10490](https://arxiv.org/abs/2307.10490)
- 62 OpenAI (2024).
Hello GPT-4o.
openai.com/index/hello-gpt-4o
- 63 OpenAI (2023).
GPT-4V(ision) system card.
openai.com/index/gpt-4v-system-card
- 64 OpenAI (2025).
Thinking with images.
openai.com/index/thinking-with-images

- 65 OpenAI (2025).
Introducing GPT-5.
openai.com/index/introducing-gpt-5
- 66 Google (2025).
A new era of intelligence with Gemini 3.
blog.google/products/gemini/gemini-3
- 67 Google (2025).
Introducing Gemini 2.5 Flash Image.
developers.googleblog.com/en/introducing-gemini-2-5-flash-image
- 68 Baltrušaitis, T., Ahuja, C., Morency, L.-P. (2019).
Multimodal Machine Learning: A Survey and Taxonomy.
IEEE TPAMI 41(2):423–443. [arXiv:1705.09406](https://arxiv.org/abs/1705.09406)
- 69 Feng, D., et al. (2021).
Deep Multi-Modal Object Detection and Semantic Segmentation for Autonomous Driving.
IEEE T-ITS 22(3):1341–1360. [arXiv:1902.07830](https://arxiv.org/abs/1902.07830)